

Pre-registration: Mating Relevant Perceptions of Voice Pitch: A Cross-cultural Examination

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Any new analyses deviated from the proposed analyses will be made transparent and reported later in the manuscript.

Pre-registration

The acoustic properties of vertebrate vocalizations are often sexually dimorphic and show evidence of having been shaped by sexual selection (Charlton & Reby, 2016; Ryan et al., 1990). Sex differences in fundamental frequency (f_0 , the rate of vocal fold vibration during phonation) derive from differences in the length and thickness of the vocal folds (Titze, 2000). Larger vocal folds with less muscular tension vibrate at a lower rate, which is perceived as lower pitch (Titze, 2000). Age-related changes in decreased f_0 have been observed in men and women; however, the pubertal rise in testosterone causes the vocal folds to enlarge and for f_0 to drop significantly in men (Fouquet et al., 2016; Harries et al., 1998; Markova et al., 2016). Consequently, compared to women, men have approximately 60% longer vocal folds and an f_0 that is five standard deviations lower (Puts et al., 2016). In many anthropoid primates, low f_0 appears to evolve in males in response to mating competition (Puts et al., 2016), perhaps because it exaggerates perceived body size (Delgado, 2006; Puts et al., 2016; Rendall et al., 2007) and hence conveys the appearance of dominance and formidability (Darwin, 1871; Puts et al., 2016). In humans, lower male f_0 has been found to increase perceptions of height, dominance, and fighting ability, suggesting that it may be a signal of formidability and/or dominance (Puts et al., 2006; Puts et al., 2007; Wolff & Puts, 2010). In female voices, higher f_0 has been found to increase perceptions of attractiveness (Borkowska & Pawlowski, 2011; Fraccaro et al., 2011; Puts et al., 2012). While many studies have shown that f_0 influences social perceptions, it is unknown (a) how variable these effects are across societies and (b) the degree to which they depend on theoretically relevant social and societal variables. The current study addresses these questions by manipulating f_0 in men's and women's voices and examining 1) mating competition-relevant perceptions of men's and women's f_0 across contexts and cultures and 2) environmental factors that may predict cross-cultural variation in perceptions of men's and women's f_0 . Below, we outline our hypotheses, method, and analysis

plans. An overview of our theoretical framework which includes main predictor variables and proposed relationships is illustrated in Figure 1.

Research Objectives

Masculinized and feminized voice preferences vary across contexts

Evidence across primates, including humans, suggests that sex differences in voice pitch likely evolved in response to the intensity of mating competition (Puts et al., 2016). In humans, male f_0 has been shown to negatively predict other men's perceptions of prestige and/or dominance and women's perceptions of attractiveness, and the effect of male f_0 appears to be stronger on dominance than prestige or attractiveness (Cheng et al., 2016; Puts et al., 2007; Saxton et al., 2016; Rosenfield et al., 2020), and stronger on attractiveness ratings for a short-term purely sexual relationship, such as a one night stand (short-term attractiveness), than a long-term committed relationship (long-term attractiveness) by women (Hodges-Simeon et al., 2011; Puts, 2005).

It is also possible that sexual dimorphism in voice pitch is due to sexual selection favoring higher pitch in women's voices as a means of mate attraction. If so, and if men generally prefer higher voice pitch in women, then women may perceive high-pitched feminine voices as indicating a greater threat in competition for mates (Puts et al. 2011; Zhang, 2016). In female voices, f_0 has been shown to positively predict other women's perceptions of flirtatiousness and attractiveness to men and men's perceptions of attractiveness (Borkowska & Pawlowski, 2011; Fraccaro et al., 2011; Puts et al., 2011; Puts et al., 2012).

In general, f_0 appears to be utilized in solving recurrent adaptive problems that would have been present in our ancestral conditions (Tooby & Cosmides, 1992; Barrett & Kurzban, 2006), especially the problems of assessing the formidability and threat potential of competitors for mates, and of choosing high-quality mates. Here, we aim to explore the functional design of f_0 in humans by examining how masculinized and feminized f_0 influences evaluations of same-sex competitors and potential mates.

Perceptions of domestic violence and danger toward children

Among women, perceived domestic violence has been shown to predict decreased preferences for masculine faces (Borras-Guevara et al., 2017a; Li et al., 2014) and voices (Li et al., 2014). Increased perceptions that men are dangerous to their children has also been shown to reduce women's facial masculinity preferences (Borras-Guevara et al., 2017b). Hence, we will test whether increased perceptions of domestic violence toward women and children reduce women's preferences for lower male f_0 on short-term and long-term attractiveness. We do not expect similar associations with male preferences for female f_0 .

Cross-cultural factors: health, inequality, food security

Given the influence of environmental variables on masculinity preferences, sex differences in perceptions of male and female f_0 may vary across societies. Previously, cross-cultural explanations in preferences for facial masculinity, but not vocal masculinity, have been reported. Findings are mixed; whereas some studies (Brooks et al., 2011; Little, DeBruine, & Jones, 2013; Scott et al., 2014) reported that both preferences and saliency of sexually dimorphic faces are strongest in societies with favorable ecological conditions (higher national health and development indices) or increased intra-sexual competition (higher homicide rates and income inequality), others (DeBruine et al., 2013; DeBruine, Jones, Little, Crawford, & Welling, 2011) suggest that female preferences for facial masculinity are stronger in nations with lower health and higher levels of pathogens. The former studies link the advantage of masculinized traits to obtaining greater material benefits and distinctive signaling toward other potential mates and rivals, whereas the latter studies link such preference to genetic benefits of preferring masculinized traits, especially in environments where offspring survival is more dependent on pathogen resistance. A recent study by Marcinkowska et al. (2019) more strongly supported the former hypothesis, reporting that female preferences for facial masculinity were stronger in nations with better health and developmental indices. We will therefore explore

relationships between societal levels of health, development, equality, and pathogen prevalence on the one hand and f_o preference on the other.

As masculinized and competitive males may be more likely to win access to resources in societies with low food security, female preference for masculinized males may be stronger in societies with lower food security and higher female social status (Brooks et al., 2011; Gleeson & Kushnick, 2018). We will therefore explore whether gender inequality interacts with food security index in predicted preferences for f_o .

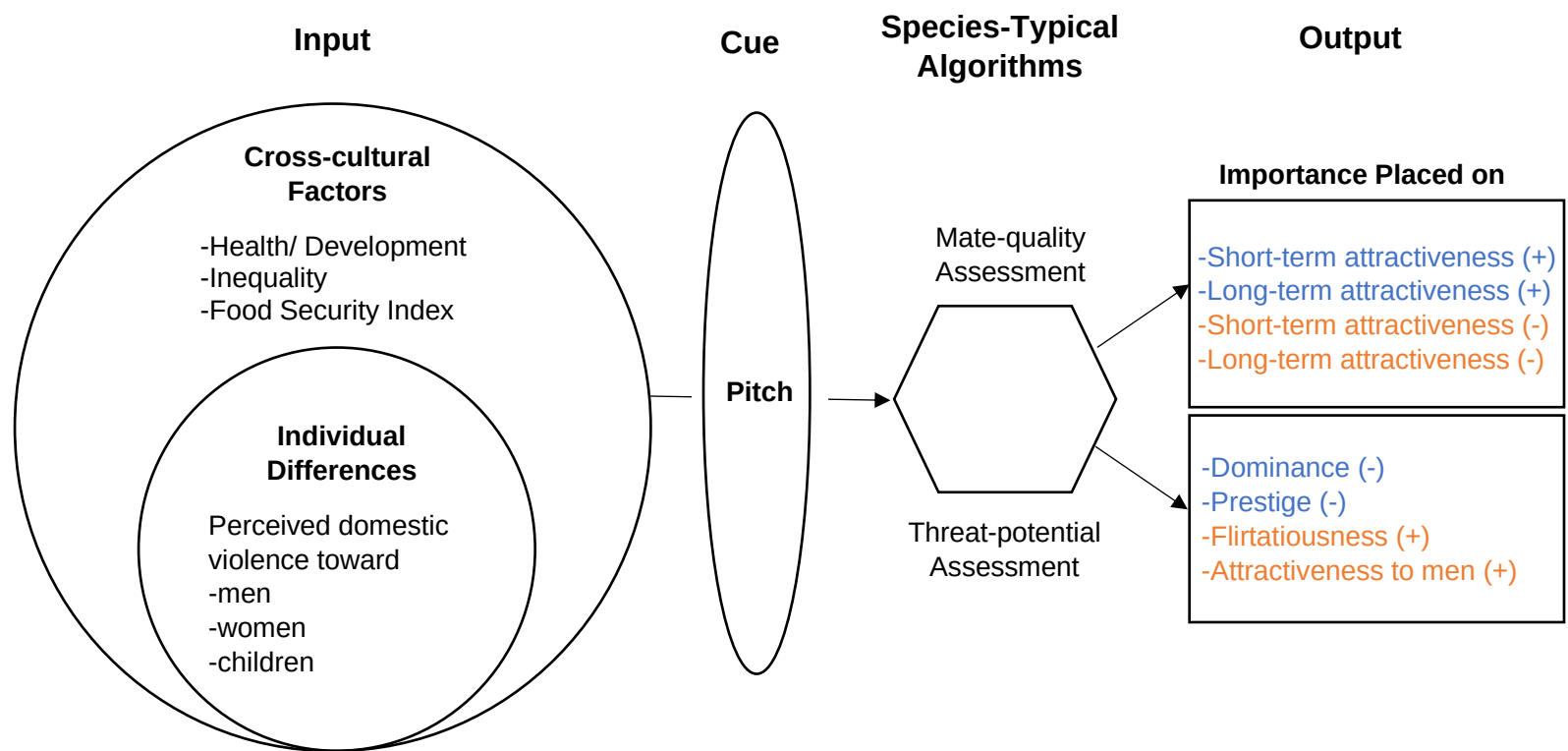


Figure 1. An overview of our theoretical framework. Inputs from cross-cultural variables and individual differences are expected to influence putative psychological adaptations in humans that utilize voice pitch as a cue for assessing mate-quality and threat-potential. Social perceptions assessed by males in our study are in blue font, and those assessed by females are in orange font. Plus (+) signs indicate expected positive relationships with pitch, and minus (-) signs indicate expected negative relationships with pitch.

Method

Participants

Data have already been collected in person from participants in 22 different countries, $n = 3,163$ (1,614 female). Participants were either from university populations ($n = 1472$) or community samples ($n = 1692$). Participants' ages ranged from 12 – 96 years old. Participants reported their relationship status; number of living children; sexual orientation (a 5-point Kinsey scale); primary language; location where they were raised, and whether they could hear the recordings clearly, understand the recordings, and hear a difference between the two voices. We aimed to collect at least 32 males and 32 females from each sampling location, but this was not possible in all cases, and in some locations many more were feasible (Fig 2).

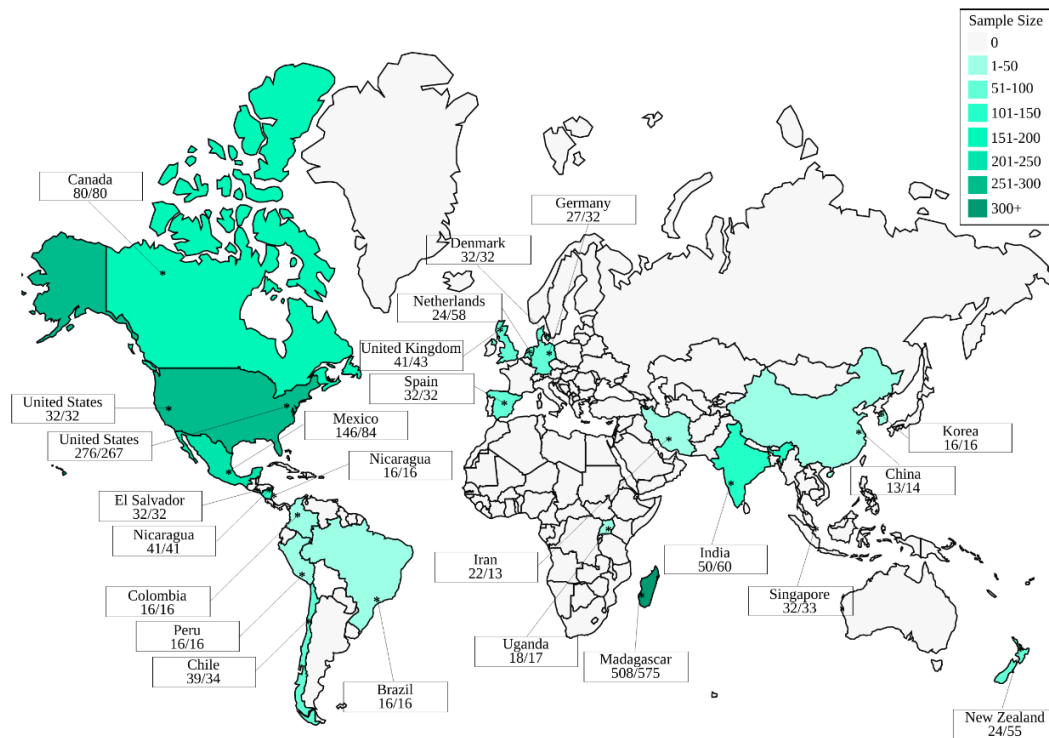


Figure 2. Locations of data collection. Each box represents sampling location and sample size broken down by sex (male/female).

Stimulus Selection, Manipulation, and Pairing

Several critical methodological decisions pertained to the selection of acoustic stimuli. In order to increase ecological validity by capturing some of the prosody of spoken language, we utilized a segment of continuous speech, rather than less natural options, such as using isolated vowel sounds. Specifically, we used the speech segment, “They act as a prism and form a rainbow,” from the first sentence of the emotionally neutral “rainbow passage” (Fairbanks, 1960). We chose this brief text in order to minimize rater fatigue, and because previous research has shown that social judgments can be made from very short voice recordings (Feinberg et al., 2005; Pisanski et al., 2012; Sorokowski et al., 2019). Another critical methodological decision pertained to whether to utilize stimuli recorded in the local languages of each sampling location, or to use the same stimuli for all raters regardless of sampling location. We opted to use identical stimuli across all participants in order to increase internal validity, and to allow us to explore possible effects of cultural factors on perceptions that were not confounded with stimulus language.

Two male and two female voice clips of the first sentence of the rainbow passage (Fairbanks, 1960) were selected from a sample of voice recordings collected at Michigan State University (Puts et al., 2016). These voices were selected because their mean f_0 was within 1SD of the mean, P_f (the average standardized formant value for the first 4 formants; Puts et al., 2012) was within 1SD of the mean, there were no errors in the first sentence of the Rainbow Passage, and they sounded natural to the experimenters when pitch was manipulated by 2SD in both directions. Additionally, only voices of women who were not taking hormonal contraception were chosen for female voice clips.

We made five pitch manipulations to each of the four voices. Using Pratt software and the means from the voice samples, each voice was manipulated to the mean pitch (Females: 5.881 ERB, Males: 3.687 ERB), to 1SD below and above the mean pitch (Females: 0.339 ERB, Males: 0.280 ERB), and to 2 SD below and above the mean pitch (Females: 0.678 ERB, Males:

0.559 ERB). Manipulation by ERB produces shifts of the same perceptual magnitude regardless of starting pitch. Unmanipulated voice clips were used as the starting point for each manipulation so that each stimulus was the result of only one manipulation. Using the software Audacity, the voice clips were cut to the 2nd phrase, "They act as a prism and form a rainbow".

Stimuli from each voice were then combined into 3 types of pairs, such that all stimulus pairs were 2SD apart: 2SD feminized pitch paired with mean pitch, 2SD masculinized pitch paired with mean pitch, and 1SD feminized pitch paired with 1SD masculinized pitch. These stimulus pairs allowed us to assess linear and curvilinear effects of voice pitch on perceptions. For example, if women prefer average male voices over deeper voices, then they should exhibit little preference when listening to 1SD feminized vs. 1SD masculinized pitch pairs, but they should tend to prefer mean pitch over 2SD feminized pitch and 2SD masculinized pitch.

Procedure

Demographic data were collected from all participants. For some countries, participants also answered 7-item questions relating to perceived vulnerability to domestic violence against men, women, and children (Borras-Guevara et al., 2017a). Participants then listened to pairs of voices and were asked to choose one voice from each pair (first or second) in answer to specific questions. Men were asked to judge male voices on which voice in each pair sounded more respected, admired, talented and successful ["prestige"], and which voice in each pair sounded more likely to win a physical fight ["dominance"]. Men were also asked to judge female voices on which voice in each pair sounded more attractive for a short-term, uncommitted romantic relationship ["short-term"], and which voice was more attractive for a long-term committed relationship such as steady dating or marriage ["long-term"]. Women were asked to judge female voices on which voice in each pair would be more attractive to men ["attractive"], and which voice in each pair sounded more interested in attracting men ["flirtatious"]. They were asked to judge male voices on which voice in each pair sounded more attractive for a short-term, uncommitted romantic relationship ["short-term"], and which voice was more attractive for

a long-term committed relationship such as steady dating or marriage [“long-term”]. The dependent variables were coded as either 0 (feminized voice chosen) or 1 (masculinized voice chosen). Each question of the 4 sex-specific questions was asked 6 times (2 voices $\times$ 3 manipulation pairs), and thus each participant rated 24 pairs of voices. Due to miscommunication, Malagasy (Madagascar) participants listened to only one randomly assigned pair of voices per question, and thus only rated 4 pairs of voices in total.

Voice stimuli were presented using iPods and Sennheiser HD280 Pro headphones in all our samples. We counterbalanced stimuli via two iTunes playlists for each sex. Each playlist featured 6 pairs of voices, with 3 pairs from each stimuli voice. Two pairs featured mean pitch first, two pairs featured masculinized pitch first, and two pairs featured feminized pitch first. The two playlists for each sex featured reversed pairs (i.e. If the 1SD masculinized version of a voice was presented before the 1SD feminized version in the first album, the feminized version would be presented first in the second album). The order of pairs of each album was randomized using the “shuffle” function on iPods. Approximately half of participants listened to album 1 of each sex and half of participants listened to album 2 of each sex.

In addition to counterbalancing pairs of voices with the use of multiple albums, male and female albums and the order of questions about the stimuli were also counterbalanced. The one exception was that male and female albums were always alternated in order to reduce rater fatigue and provide greater independence between responses to the two questions about each sex of speaker. That is, no participant listened and answered questions about one sex’s voices twice in a row — participants either heard albums in the order of male-female-male-female or female-male-female-male. This created a total of 16 different respondent sheets for each sex in addition to randomized order of the presentation of stimuli.

Cross-cultural Variables

Participants might prefer voices closer to the local average pitch. To test this possibility, we aim to collect voice recordings from 20 participants (10 males and 10 females), speaking

short phrases in the official language of each country where data were collected. These voice recordings will be primarily obtained from the localingual website (Ding, 2017), an interactive platform where participants from various countries can voluntarily contribute their audio recordings. Using Praat software, mean pitch of each sex will be extracted across individuals for each country and used as a predictor.

This study will also extract unique factor levels from various inter-correlated country-level measures (see Table 1). Following Marcinkowska et al. (2019), we will conduct independent factor analysis and use these extracted measures, as well as the food security index (Economist Intelligence Unit, 2019), to predict preferences for vocal masculinity (lower f_0) across questions on various mating relevant traits. The mean statistic in our sample will be used as a replacement for missing data of each variable. As in Marcinkowska et al. (2019), we expected to obtain two extracted variables: health/development & Inequality factors.

Table 1. Data source and collected cross-national variables

Data Source	Variables
World Bank Database	Women's fertility rate, Homicide rate, Proportion of the population urbanized, Gross domestic product, Mortality rate, Gini coefficient
The World Health Organization	Years lost to communicable disease and adult life expectancy
The United Nations	Human Developmental Index, Gender Inequality Index
Murray & Schaller (2010)	Historical disease prevalence

Analysis Plan

Analyses will be conducted using Bayesian multilevel models via the statistical software R (R Core Team, 2016), ape package (Paradis E. & Schliep, 2018), and brms package (Bürkner, 2017). As mating competition is most likely to be relevant to the ages between 16 and 40, we limit our analyses to this age range during hypothesis testing. Main hypotheses will be tested using logistic mixed effects models, where choices (either 0 or 1) will be entered as

dependent variables. To test whether participants' preferences for masculinized and feminized voices across contexts, vocalizer sex (male, or female voices), question types (e.g., prestige, dominant, short-term, long-term), and stimulus types (3 stimulus pairs) will be entered as fixed effects. In a separate model, participants' age, relationship status, number of children, and sexual orientation will be entered as fixed effects to test whether participants' demographic characteristics influence their preferences for masculinized and feminized voices across questions. To test whether perceived domestic violence risk shapes men's and women's voice preferences for the opposite sex in short-term and long-term mating contexts, we will run two separate models where question types, stimulus types, and perceived domestic violence ratings toward women, men, and children will be entered as fixed effects. As in Marcinkowska et al. (2019), independent factor analysis will be used to extract unique country-level factors to account for highly inter-correlated cross-cultural variables and we will use these extracted factors, as well as the mean pitch of the local language, to predict preferences for masculinized and feminized voices across each question in our study. Across all models, the potential non-independence of observations made by the same participant, participants in the same country, and the use of same stimuli across participants will be accounted for by including participant ID, country, and voice ID as random intercepts. Geographical region will be included as an additional random effect to account for potential non-independence between countries (e.g., similar climate, shared cultural history, etc.). In addition to geographical relatedness between nations, we will also control for the phylogenetic relatedness between nations using linguistic distance matrix. Random slopes will be specified maximally, as recommended by (Barr et al., 2013).

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